

THESIS

A STUDY OF COLLABORATIVE FUNDING FROM THE NATIONAL INSTITUTES OF HEALTH
FROM SOCIAL NETWORK PERSPECTIVES

Submitted by

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ABSTRACT

A STUDY OF COLLABORATIVE FUNDING FROM THE NATIONAL INSTITUTES OF HEALTH FROM SOCIAL NETWORK PERSPECTIVES

In the past decade, academic collaboration between organizations has been increasing. The believed benefits include access to more ideas, information and skills than a single organization can produce on its own. In this thesis, we analyze funding data extracted from the National Institutes of Health (NIH). We adopt social network approaches to investigate relationships between collaborating organizations. In particular, organizations are modeled as nodes in the network, and an edge, representing the number of collaborations, connects each pair of nodes. We model the network in four different frameworks to explore the importance of primary-secondary organizational relationships and repeated collaborations. We highlight important and influential organizations in the network with respect to various topological descriptive statistics. This work will provide insights into the landscape of collaborative funding and, potentially, the diffusion pathways of innovative research.

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1 Introduction

The National Institutes of Health (NIH) is the world’s leading financier of medical related research, funding over 2,500 research organizations including more than 300,000 research personnel [1]. Recently, the question has been raised about which organizations receiving this funding collaborate in order to more effectively engage in and answer challenging research inquiries. Collaborations between different organizations is becoming more popular among universities receiving grants funded by the National Science Foundation [2]. These collaborations provide an advantage for all parties involved because no two organizations are alike and thus have different skill sets and abilities [3]. A project that is being completed by two or more organizations simultaneously has greater access to information and these different skills, which is crucial for conducting cutting edge research [4]. Beyond this, some research projects are becoming increasingly complex, so larger and more diverse teams are needed to answer the questions at hand [5, 6].

In this paper, we will focus on analyzing a dataset consisting of different organizations that have received a collaborative grant that was being funded between 2001 and 2014. For our purposes, a collaborative grant is one that is defined as follows: a grant that funds a research project such that the project title is shared by at least two distinct organizations and has at least two distinct principal investigators involved. Since these projects were performed by principal investigators from different organizations, it is reasonable to assume that collaboration had taken place [7]. There may be some problems with measuring collaboration in this fashion [8], but it is still one of the most common ways to do so [9]. The analysis of collaboration in this paper is limited to research grants as labeled by the NIH. The vast majority of grants funded by the NIH were non-collaborative as defined above, but 1,200, about 0.74%, research grants were collaborative.

2 NIH Data Description

The data used for this paper are publicly available from the NIH website. We accessed these data using the ExPORTER tool also available on the NIH website [10]. The database consists of nearly 340,000 distinct project titles funded by the National Institutes of Health between the years of 2001 and 2014 with around 167,000 of these being classified as research projects. For each entry in the database, the available information includes project title, principal investigator, organization, the year in which the project was funded and award amount (funding) received. If the same project was conducted by multiple organizations, the project title is entered into the database multiple times, once for each organization. For each funded project, different entries were created for each year as well.

In theory, identifying projects that are collaborative is simple. If the same project title with more than one principal investigator has more than one organization with which it is associated, it is collaborative. In practice, however, it is rather complicated. There appears to be no specific and systematic way to enter each organization’s name into the database, which resulted in inconsistencies in the names of organizations in the database. For example, the Bozeman campus of Montana State University was entered as `Montana State University - Bozeman` for some project titles and listed as `Montana State University (Bozeman)` for others. Some upfront data cleaning was needed because of these inconsistencies. This also occurs, although to a lesser extent, with the projects; some project titles have slightly different names in the database.

To identify collaborative grants, we started with a list of organizations that were entered in the database at least once per year in at least six of the fourteen years (2001 to 2014). This could mean it had six different one-year grants, one six-year grant or anything in between. We excluded some organizations, like businesses and corporations, since our focus is directed toward analyzing networks related to academic and health organizations. This gave us a starting list of around 700 organizations. The next step was to find each collaborative grant

of which these organizations were a part, obtain the name of the project this grant funded and then keep track of which organizations were involved. We did not consider projects related to administration, data analysis, or other such topics, but instead focused on research projects. This gave us a set of 1,230 projects and 456 organizations. Note that many of the original organizations were eliminated since they were not all involved with collaborative grants.

3 Network Basics

To effectively analyze the NIH funding dataset, we consider modeling such data as a large network. Here, we will briefly review basic social network structures.

A *network* is defined by a set of nodes (or vertices) connected by edges (or ties). Each organization in our organization list is represented by a node and is connected to another organization by an edge if collaboration between them is identified. A network can be defined as directed or undirected and weighted or unweighted based on certain properties or attributes of edges.

The edges in a directed network have an ordering that the edges in an undirected network do not have. In our problem, the edges in directed networks extend out from the primary organization, which will be defined soon, to the secondary organization. The maximum number of edges in directed networks is twice as large as the maximum number of edges in an undirected network since there could be an edge from node i to node j and from node j to node i .

The weighted networks will have edge weights associated with one or more attributes of the network. For the NIH network, the edges will be weighted by the number of collaborations between organizations, i.e., the number of grants shared by two organizations. The four network representations that will be used to model the NIH network are the four combinations of undirected/directed and unweighted/weighted.

For each collaboration, the organization that worked on it the most number of years was determined to be the primary organization for that project. If there was more than one organization that had worked on the project for the same (maximum) number of years, the organization that had worked on it first was deemed to be the primary one. If multiple organizations had worked on the project for the same number of years and had all been involved during the initial year, then that project would have multiple primary organizations. This was the case in 142 of the 1,230 projects, which comprises about 11.5% of all collaborative projects. All organizations associated with a project that were not considered primary are going to be referred to as secondary organizations.

This notion of primary organizations was used when creating the directed networks. For undirected network representations, any organizations, the nodes, that worked on the same project will be connected with an edge. When dealing with directed network representations, we say each non-primary or secondary organization only worked with the primary organization. Therefore, we modeled the information flow as being from the primary organization to the secondary organizations, with no information traveling between secondary organizations.

The distributions of edge weights are given in Figure 1. In the undirected, weighted network representation, 78% of the 3,082 edge weights are equal to one with a maximum weight value of eighteen and in the directed, weighted representation, 86% of the 2,280 edge weights are equal to one with a maximum of seventeen. Although many edges only have a weight of one, which is to say that most edges are not at all different from their unweighted counterparts, the weighted representations differ enough to warrant a separate analysis.

Mathematically, a network can be represented by an adjacency matrix. This adjacency matrix is coded differently for all of the network representations. For undirected networks, entry (i, j) in the matrix will be the number of grants organization i and organization j have worked on together in some capacity. This number is determined by whether organizations i and j were ever listed under the same project title. Undirected adjacency matrices are

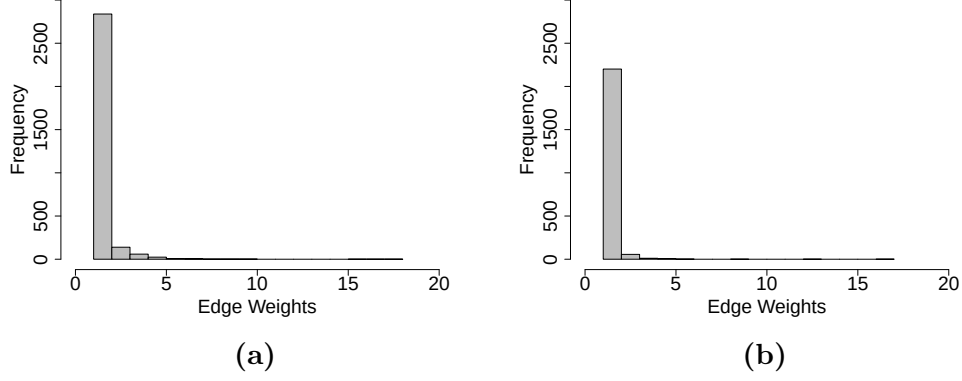


Figure 1: Distributions of Edge Weights in the weighted networks. (a) Distribution in undirected, weighted network representation. (b) Distribution in directed, weighted network representation.

symmetrical so that entry (i, j) equals entry (j, i) . If two organizations have not worked together, the corresponding entries will be zero.

Directed adjacency matrices are usually not symmetric. The entry (i, j) represents the total number of projects that both organizations i and j were involved in where organization i was the primary. It is worth mentioning that the research projects with both organizations listed as primary are included in both entries (i, j) and (j, i) .

The adjacency matrices for the unweighted networks are the thresholded versions of the weighted adjacency matrices. That is to say, one can simply change any nonzero entry in the weighted adjacency matrices to a one to create the adjacency matrix for unweighted networks.

We will denote the undirected, weighted adjacency matrix, also called the undirected weight matrix, $W = [w_{ij}]_{i,j=1}^N$ where N is the number of nodes in the network, which, in this case, is 456. The directed weight matrix is $W^D = [w_{ij}^D]_{i,j=1}^N$. We will also make a distinction between the individual elements of the weight matrix and the thresholded elements of that matrix. The thresholded elements are denoted a_{ij} in the undirected case and a_{ij}^D in the

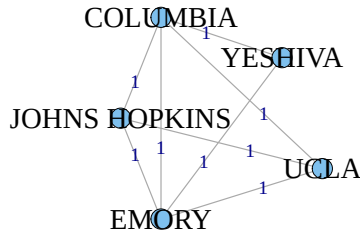
directed case. So

$$a_{ij} = \begin{cases} 1 & \text{if } w_{ij} > 0 \\ 0 & \text{if } w_{ij} = 0 \end{cases}$$

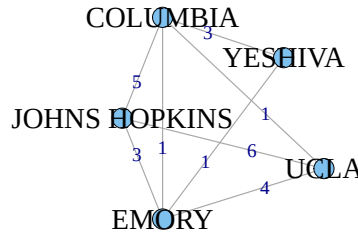
and

$$a_{ij}^D = \begin{cases} 1 & \text{if } w_{ij}^D > 0 \\ 0 & \text{if } w_{ij}^D = 0 \end{cases}$$

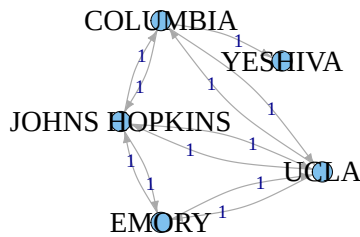
Then the undirected, unweighted adjacency matrix is the thresholded undirected, weighted adjacency matrix, which will be denoted $A = [a_{ij}]_{i,j=1}^N$ and the directed, unweighted adjacency matrix is the thresholded directed, weighted adjacency matrix, which will be denoted $A^D = [a_{ij}^D]_{i,j=1}^N$.



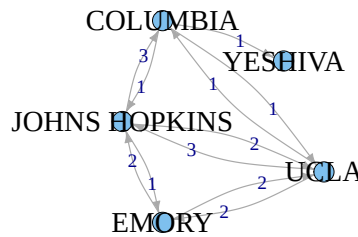
(a)



(b)



(c)



(d)

Figure 2: Examples of Different Network Representations. The edge labels correspond to the weights of each edge. (a) Undirected, unweighted network representation. (b) Undirected, weighted network representation. (c) Directed, unweighted network representation. (d) Directed, weighted network representation.

To illustrate these different network representations, we look at a subset of the NIH net-

work. Figure 2 shows this subset for each of the four network representations we will be examining with only five organizations (nodes) and all edges connecting them. Figure 2a shows the undirected and unweighted network representation. This tells us which organizations have worked together at least one time. For instance, Emory University has worked with all of the other organizations at least once, but the University of California Los Angeles (UCLA) and Yeshiva University have never worked on the same project. Figure 2b expands on the information given in Figure 2a. In particular, we can see that Johns Hopkins University and UCLA have been involved with six projects, while UCLA and Columbia University Health Sciences have only worked together once.

Next, Figures 2c and 2d tell a completely different story because they are representations of directed networks. Figure 2c indicates that Johns Hopkins University was the primary organization in at least one project that included UCLA, Emory University and Columbia University Health Sciences since there is an edge extending from Johns Hopkins to those other schools. Yeshiva University had never been a primary organization of a grant. Figure 2d then expands this to show that Johns Hopkins University was the primary organization of two different projects in which UCLA was a secondary organization. UCLA has been the primary organization of three projects with which Johns Hopkins University was involved.

These graphical illustrations display different information we can obtain when evaluating these four different networks. Depending on what research question is being asked, any of these networks could be useful, as we will observe later in this paper.

4 Network Data Visualization

The main tools used to visually display the NIH networks were 1) the **igraph** package [11] within **R**, an open source and very popular statistical software package [12], and 2) an open

source software **Gephi**¹ [13]. Node size is adjusted by out-degree in the unweighted networks and by weighted out-degree in the weighted networks. Degree will be defined in Section 5.1.

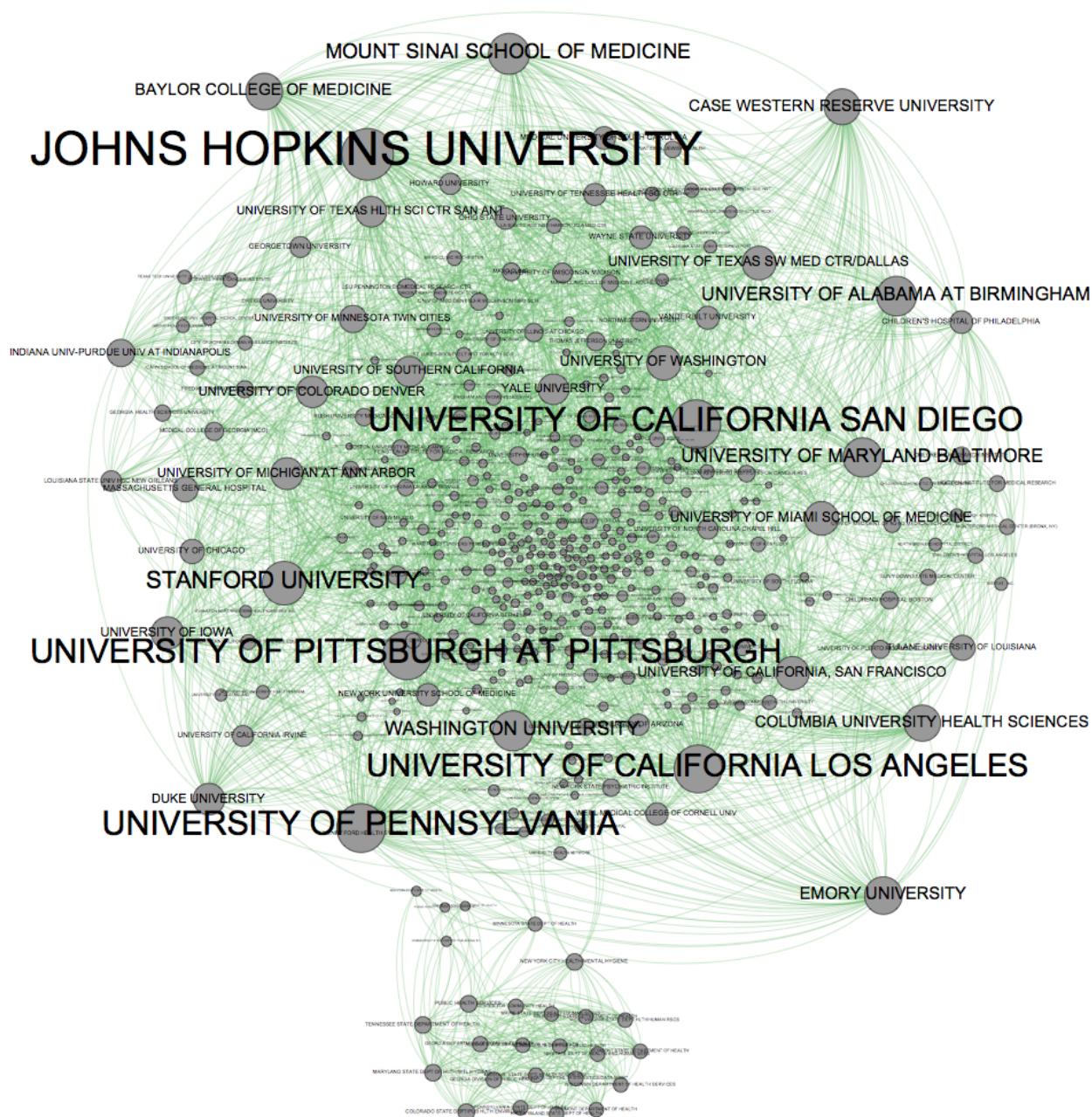


Figure 3: Undirected, Unweighted NIH Network Representation. Node size and node label are adjusted based on node degree.

¹The visualization method used in Gephi is called ForceAtlas with parameters as follows: inertia = 0.4, repulsion = 200, attraction = 25, maximum displacement = 5, autostabilize = yes, autostab strength = 80, autostab sensibility = 0.3, gravity = 1000, attraction distribution = yes, adjust by size = yes

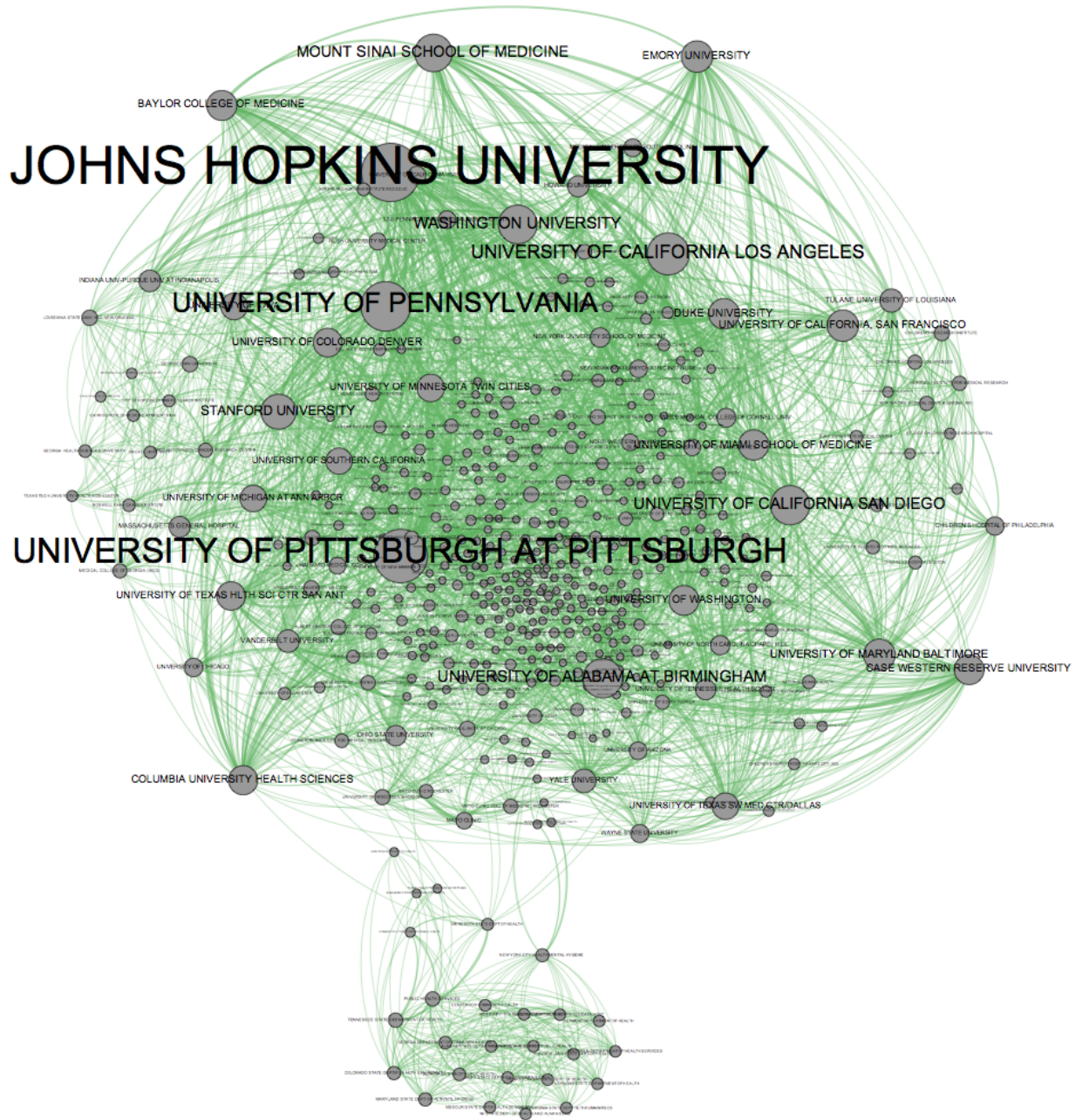


Figure 4: Undirected, Weighted NIH Network Representation. Node size and node label are adjusted based on node strength.

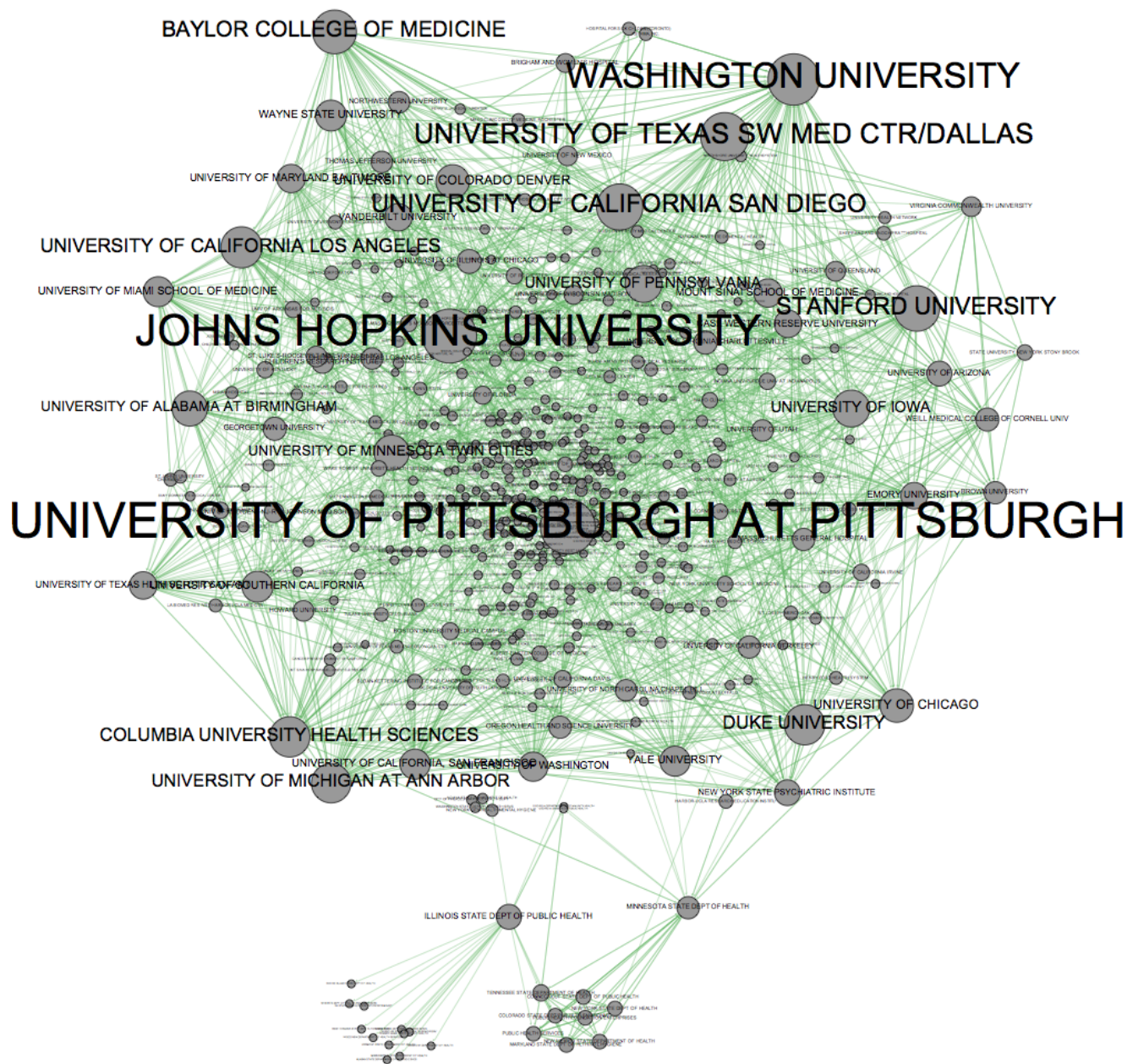


Figure 5: Directed, Unweighted NIH Network Representation. Node size and node label are adjusted based on node out-degree.

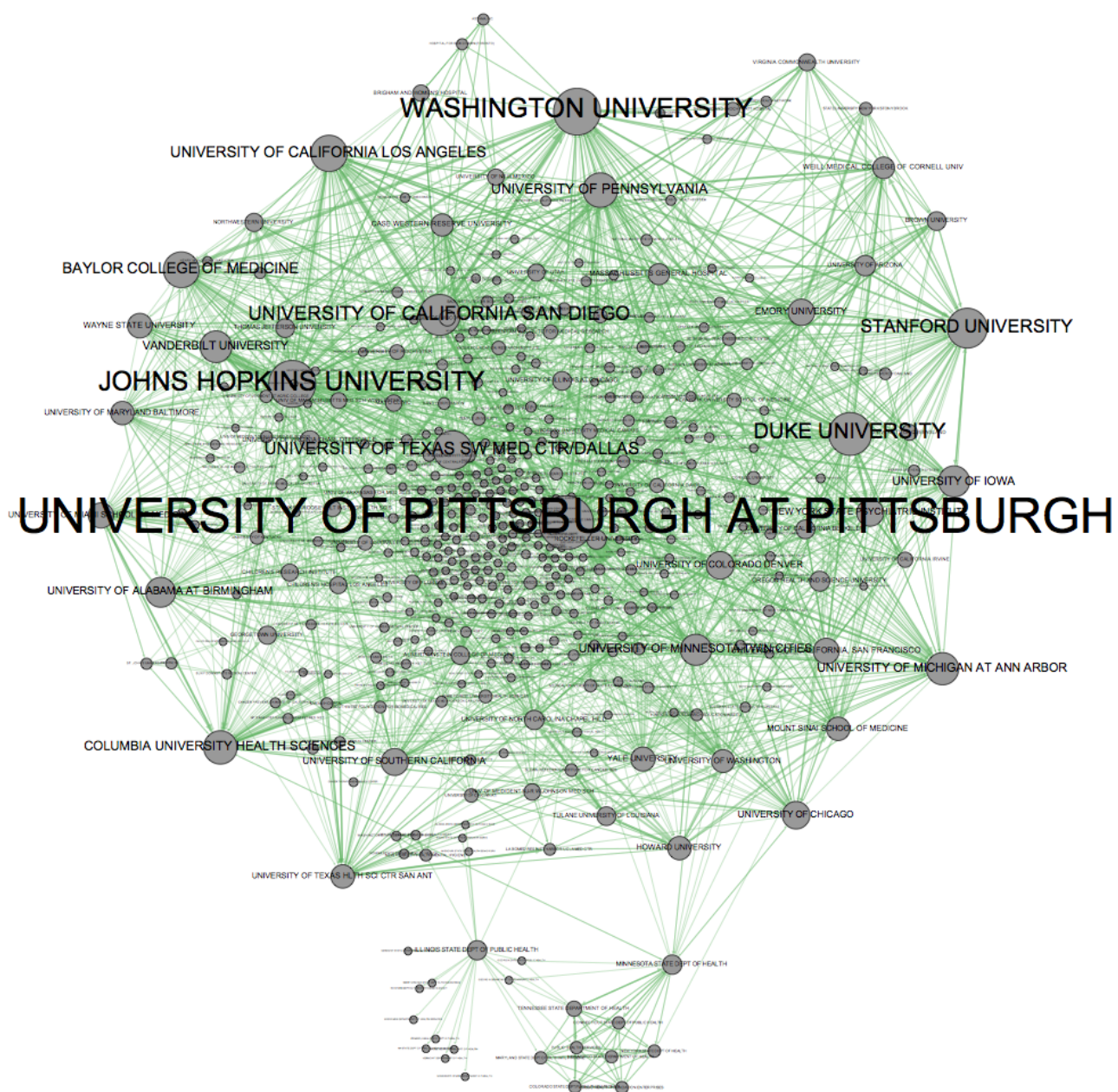


Figure 6: Directed, Weighted NIH Network Representation. Node size and node label are adjusted based on node out-degree.

Figure 3 shows the undirected, unweighted NIH network representation. Figure 4 shows the undirected and weighted NIH network representation. Figure 5 shows the directed, unweighted NIH network representation and finally Figure 6 shows the directed, weighted

NIH network representation. From all four of these visualizations, it is clear that the NIH network is divided. On the bottom, we see a smaller network that is quite interconnected but is mostly connected with the rest of the network above through just a few nodes. Those nodes that connect these networks are Minnesota State Department of Health and Illinois State Department of Health. The smaller network is mostly made up of other state departments of health.

5 Network Descriptive Statistics

The main goal of analyzing these four network representations is to calculate descriptive statistics for each and compare them. There are many different ways to describe a network, but some of the most commonly used and useful descriptive statistics include the average network degree, the clustering coefficient and the average shortest path. We will also look at a few other statistics including closeness centrality, betweenness centrality, eigenvector centrality, reciprocity and altruism level.

5.1 Network Degree

Network degree is the number of connections each node has to other nodes in the network. This will quantify the connectedness of each organization. For a funding network, a node (an organization) with more connections is usually regarded as better since this means the organization is involved with more projects and thus is learning from and teaching others, building collaborations and tackling more projects. In an undirected and unweighted network, this is as simple as summing the row totals of an adjacency matrix. Therefore, the degree of a node is

$$k_i = \sum_{j=1}^N a_{ij}, \quad (1)$$

where N is the number of nodes in the network, which, again, is 456 in our case, and a_{ij} is the entry on the i th row and the j th column of the adjacency matrix, denoted by A .

For a directed network, there are three distinct, inter-related degree measures. The *out-degree* for node i is the number of edges originating at node i and extending out to other nodes. The *in-degree* for node i is the number of edges originating from other nodes and terminating at node i . Finally, the *total degree* of a node is the sum of both out-degree and in-degree. We will be more interested in the out-degree for directed networks since it will show us how many times each organization has been the primary one when working on a project. Out-degree for a directed, unweighted network can be calculated exactly the same way as is shown in Equation (1). We can simply replace the elements of A with the elements of A^D :

$$k_i^D = \sum_{j=1}^N a_{ij}^D$$

To find the in-degree in a directed network, the columns of the adjacency matrix, rather than the rows, are summed. The in-degree for node i will be denoted as l_i^D .

$$l_i^D = \sum_{j=1}^N a_{ji}^D$$

In undirected networks, the out-degree is equal to the total degree but, of course, it does not tell us which organization is primary.

Extending the idea of network degree to weighted networks, we use weighted degree, also known as node strength, as a substitute [14]. Node strength, both for directed and undirected networks, is found by summing the rows of the weight matrix:

$$s_i = \sum_{j=1}^N w_{ij}$$

where w_{ij} is the entry in the i th row and the j th column of the weight matrix, W . Again,

simply replace the elements of W with the elements of W^D to obtain the node strength in the directed, weighted network:

$$s_i^D = \sum_{j=1}^N w_{ij}^D$$

5.2 Clustering Coefficient

The mean local clustering coefficient is the proportion of possible connections in the neighborhood of node i that are connected. A neighborhood in this context contains all of the nodes to which node i is connected, including any possible connections between nodes to which node i is connected. The clustering coefficient can be thought of as a measure of collaboration. Essentially, it gives the proportion of the friends of a certain node being friends themselves. This is measured by examining the number of complete triangles in the network in relation to the total number of possible triangles. Figure 7 provides a graphical display of a triangle in a network of three nodes. Figure 7a shows a complete triangle and Figure 7b shows an incomplete triangle.

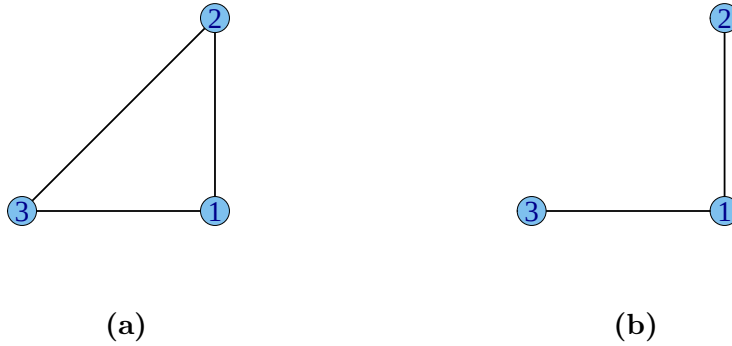


Figure 7: Triangles in Networks. (a) A complete triangle. All three nodes are connected to one another. (b) An incomplete triangle. Here, node 1 is connected to the others, but nodes 2 and 3 are not connected themselves.

In Figure 7a, all three nodes form one triangle. For node 1, $t_1 = 1, k_1 = 2$ and so, using Equation (2), the local clustering coefficient for node 1 is $C_1 = \frac{2 \cdot 1}{2(2-1)} = 1$. This makes sense if one thinks of the local clustering coefficient as the proportion of the friends of a certain

node being friends themselves. Since all the connections node 1 has (nodes 2 and 3) are also connected themselves, the proportion of the friends of node 1 that are friends themselves is one. Figure 7b shows an incomplete triangle. While node 1 is connected to both node 2 and node 3, node 2 and 3 are not connected to one another. So the local clustering coefficient for all three nodes will be zero since they are all not a part of a triangle.

We will have a distinct clustering coefficient for each node, each called a local clustering coefficient. The clustering coefficient is a measure of the transitivity of a network and is also known simply as transitivity. We have four different ways to calculate the clustering coefficient: one for each network representation.

5.2.1 Undirected, Unweighted Clustering Coefficient

First, we will look at the local clustering coefficient [15] for the undirected and unweighted network. This is denoted as C_i and is given by

$$C_i = \frac{2t_i}{k_i(k_i - 1)} = \frac{(A^3)_{ii}}{k_i(k_i - 1)} \quad (2)$$

where t_i denotes the number of triangles of which node i is a part, k_i is the degree of node i and A is the adjacency matrix for the network. Here, $(A^m)_{ii}$ is a walk of length m around node i . When $m = 3$, this gives twice the number of triangles for node i . The triangles are double counted since A can be traversed in two directions. Then the mean local clustering coefficient, which measures the overall level of clustering in the network, is simply $C = \frac{1}{N} \sum_{i=1}^N C_i$. Also, if $k_i = 1$ for any i in the network, then C_i is defined to be equal to zero.

5.2.2 Undirected, Weighted Clustering Coefficient

The clustering coefficient is defined differently for undirected, weighted networks. There have been many different clustering coefficients proposed for weighted networks, but a commonly used one is based on the work of Barrat et al [14]. Their clustering coefficient is given by

$$\tilde{C}_i = \frac{1}{k_i(k_i - 1)} \sum_{j,k} \frac{1}{\langle w_i \rangle} \frac{w_{ij} + w_{ik}}{2} a_{ij} a_{jk} a_{ik},$$

where $\langle w_i \rangle = \sum_j w_{ij}/k_i$. Essentially, the numerator in Equation (2) is replaced by weighted triangles. Each triangle that node i is a part of is weighted by the ratio of the average weights of the adjacent sides of the triangle to the overall average weights of all triangles of which node i is a part. Define the average clustering coefficient for undirected, weighted networks to be $\tilde{C} = \frac{1}{N} \sum_{i=1}^N \tilde{C}_i$.

5.2.3 Directed, Unweighted Clustering Coefficient

Now, we have two more measures of transitivity: one for the directed, unweighted network and one for the directed, weighted network. Fagiolo proposed the following two methods to measure transitivity for directed networks [16]. For the directed, unweighted network, the transitivity is defined as “the ratio between all directed triangles actually formed by i (t_i^D) and the number of all possible triangles that i could form (T_i^D)”. This can be written

$$C_i^D = \frac{t_i^D}{T_i^D} = \frac{\left(A^D + [A^D]^T \right)_{ii}^3}{2 \left[k_i^{D,tot} (k_i^{D,tot} - 1) - 2k_i^{D,\leftrightarrow} \right]}$$

where A^D is the directed adjacency matrix, $k_i^{D,tot}$ is the total degree (in and out) for node i in the directed network, and $k_i^{D,\leftrightarrow} = \sum_{j \neq i} a_{ij} a_{ji} = [A^D]_{ii}^2$, which gives us the number of reciprocated edges between node i and its neighbors. Again, $C^D = \frac{1}{N} \sum_{i=1}^N C_i^D$.

5.2.4 Directed, Weighted Clustering Coefficient

Finally, we will discuss the transitivity for directed, weighted network. This transitivity is given by

$$\tilde{C}_i^D = \frac{\tilde{t}_i^D}{T_i^D} = \frac{\left(W^D[\frac{1}{3}] + \left[W^D[\frac{1}{3}] \right]^T \right)_{ii}^3}{2 \left[k_i^{D,tot} (k_i^{D,tot} - 1) - 2k_i^{D,\leftrightarrow} \right]}$$

where W^D is the directed weight matrix and $W^D[\frac{1}{3}]$ is the cube root of each entry in W^D to prevent the weights from being counted thrice. The overall level of clustering for the weighted, directed network is $\tilde{C}^D = \frac{1}{N} \sum_{i=1}^N \tilde{C}_i^D$.

5.3 Average Shortest Path

The shortest path measure provides a way to quantify how connected a network is [17]. For each node, the shortest distance to each other node in the network is calculated by counting how many edges it needs to pass over before arriving at its destination. This can be easily extended to weighted networks by associating a “cost” for passing through each edge equal to the weight of that edge. However, we will calculate the shortest path for weighted networks with the inverse of the weights, as proposed by Newman [18], since we think of it as being easier to go through one of the nodes with a strong connection with the starting node than through a node that has a weak connection.

In the directed network context, we will only allow the path to follow the direction of that edge. For example, if an edge is directed from node i to node j , then the path from i to j is equal to one while the path from j to i is equal to infinity unless there is another path from node j to node i . For all networks, the average shortest path statistic will be the mean of the shortest paths for each node to each other node, excluding the nodes that are disconnected (those that have no complete path to one another). The shortest path from

node i to node j will be written as $d(i, j)$ and the average shortest path will be denoted

$$\bar{d} = \frac{1}{N_d} \sum_{i,j} [d(i, j) \cdot I_{d(i,j) \neq \infty}]$$

where $I_{d(i,j) \neq \infty}$ is the indicator function that is equal to one when $d(i, j) \neq \infty$ and zero otherwise, and N_d is the number of shortest paths in the network that are not equal to zero or infinity. It should be noted that there are only six organizations that are not connected to the rest of the network in the undirected network representations. Therefore, the paths that equal infinity are only when trying to connect other nodes to these six. In the directed network representations, that number jumps up to thirty-two. Even in the directed case, this only accounts for 7% of all nodes, so ignoring the paths that equal infinity will not skew the results too much.

To further illustrate how shortest paths change in directed and weighted networks, we turn to Figure 8. Here we are looking at the shortest path from Emory University to Yeshiva University. Figure 8a shows the shortest path in the undirected, unweighed network representation. Emory and Yeshiva are directly connected and so the path length is equal to one since only an edge with a weight of one (the weight of all edges in an unweighted network) was crossed. Figure 8b shows how the shortest path changes in the undirected, weighted network representation. Since we use the inverse weights as edge weights when finding shortest paths, the shortest path between Emory and Yeshiva is now the path through Johns Hopkins University and Columbia University Health Sciences. This is because Emory and Johns Hopkins worked on three projects together, Johns Hopkins and Columbia worked on five projects together and Columbia and Yeshiva worked on three projects together. Then, the cost associated with traveling from Emory and Johns Hopkins is $1/3 \approx 0.333$; the cost associated with traveling from Johns Hopkins to Columbia is $1/5 = 0.2$; and the cost associated with traveling from Columbia to Yeshiva is $1/3 \approx 0.333$. Therefore, the total cost from Emory to Yeshiva when going through Johns Hopkins and Columbia is 0.867, whereas

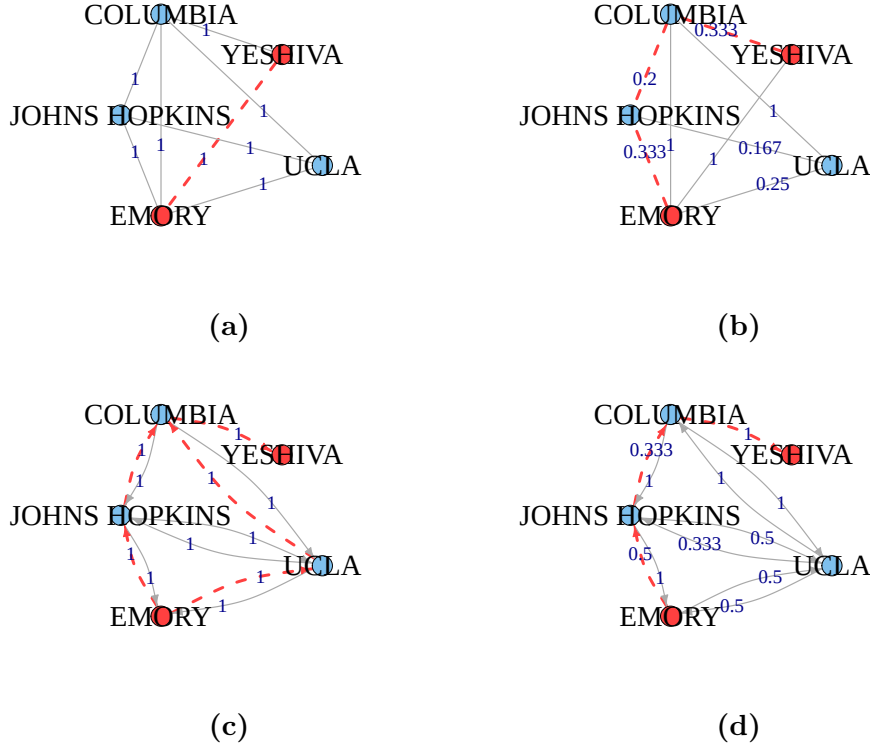


Figure 8: Shortest Paths. The edge labels correspond to edge weights and the bold dotted path represents the shortest path from Emory University to Yeshiva University. (a) The shortest path in the undirected, unweighted network representation. (b) The shortest path in the undirected, weighted network representation. (c) The shortest path in the directed, unweighted network representation. There are two shortest paths in this case. (d) The shortest path in the directed, weighted network representation.

the cost of going directly to Yeshiva from Emory is equal to one. So we will say the shortest path length from Emory to Yeshiva in the undirected, weighted network representation is equal to 0.867.

Now, Figure 8c shows the shortest paths from Emory to Yeshiva in the directed, unweighted network representation. Notice that Emory and Yeshiva are no longer connected in this directed network and neither is Emory connected to Columbia. Thus, there are actually two shortest paths from Emory to Yeshiva, both of length three. The first path goes from Emory University → Johns Hopkins University → Columbia University Health Sciences → Yeshiva University. The second replaces Johns Hopkins with UCLA.

In the directed, weighted network representation, there is again one clear shortest path. The cost associated with going through UCLA and Columbia is now equal to 2.5 when adding the inverse weights to the graph. The cost of going through Johns Hopkins and Columbia is now equal to 1.833. Therefore, the shortest path will change depending on which network representation we are referencing. However, it should be noted that the weighted network will always have an average shortest path length that is less than or equal to that of its unweighted counterpart.

5.4 Closeness Centrality

Both closeness and betweenness were explained by Freeman [19] as measures of node centrality, which is a measure of how crucial each node is in the network. As the name implies, it measures how close a node is from other nodes in the network, similar to shortest path. Closeness for node i is measured by summing the shortest path distances between node i and all other nodes and then taking the inverse of that sum. We take the inverse so that nodes that are far from other nodes have a small closeness measure and then multiply that value by $N - 1$ to allow it to be compared to networks with a different number of nodes. Mathematically, the formula is

$$Cl_i = \frac{N - 1}{\sum_{j=1}^N d(i, j)}$$

where $d(i, j)$ is the shortest path distance between node i and j . For this measurement, if any node is disconnected from any other node (the length is infinity) that shortest path distance will be replaced by the total number of nodes in the network, which is 456. Closeness is easily extended to weighted and directed networks by replacing $d(i, j)$ with each respective shortest path distance.

5.5 Betweenness Centrality

Betweenness measures the frequency that node i falls between nodes j and k when connecting nodes j and k with a shortest path. Another way to think of betweenness for node i is as the number of shortest paths going through node i . It is given by

$$B_i = \sum_{j < k} \frac{g_{jk}(i)}{g_{jk}} \quad \text{for } i \neq j \neq k$$

where g_{jk} is the number of shortest paths from node j to node k and $g_{jk}(i)$ is the number of these shortest paths that pass through node i .

We will then normalize this quantity by dividing by the largest possible number that it can be, which is $\frac{(n-1)(n-2)}{2}$ in undirected graphs and $(n-1)(n-2)$ in directed ones. This normalization is done so that the betweenness values can be directly compared. Thus, the standardized values for undirected and directed graphs are given by

$$B'_i = \frac{2}{(n-1)(n-2)} \sum_{j < k} \frac{g_{jk}(i)}{g_{jk}} \quad \text{for } i \neq j \neq k$$

and

$$B_i^{D'} = \frac{1}{(n-1)(n-2)} \sum_{j < k} \frac{g_{jk}(i)}{g_{jk}} \quad \text{for } i \neq j \neq k$$

5.6 Eigenvector Centrality

Eigenvector centrality, explained by Jackson [20], is a measure of how well connected a node is to other well connected nodes. The values for this centrality measure are between zero and one with one being the highest (connected to many other well connected nodes). As the name implies, these values are calculated by taking the eigenvector that corresponds to the largest eigenvalue in the adjacency matrix and scaling it so that the maximum value is one.

The usual definition of an eigenvector applies:

$$\lambda C_i^e = AC_i^e$$

where C_i^e is the eigenvector centrality, λ is the largest eigenvalue of A , and A is an adjacency matrix. This same definition holds for directed and weighted adjacency matrices as well by simply replacing A with the adjacency matrix for the network in question.

5.7 Reciprocity

Reciprocity is a helpful tool when analyzing directed networks [21]. Reciprocity measures the proportion of edges in the directed network that are bidirectional. To obtain this statistic for the directed, unweighted network, the number of edges in the network that point both ways is divided by the total number of edges:

$$r = \frac{E^{\leftrightarrow}}{E} = \frac{\sum_{i,j=1}^N a_{ij}^D a_{ji}^D}{\sum_{i,j=1}^N a_{ij}^D}$$

where $E^{\leftrightarrow}$ is the number of edges that point in both directions, E is the total number of edges in the network and a_{ij}^D is the element in the i th row and j th column of the directed, unweighted adjacency matrix. This can be thought of as organizational reciprocity. It shows how many organizations are both primary and secondary organizations with some other organization in the network.

When extending this idea of reciprocity to weighted networks, reciprocity is computed differently. We will use the weighted reciprocity as proposed by Squartini et al. [22]:

$$r_w = \frac{\sum_{i,j=1}^N \min(w_{ij}^D, w_{ji}^D)}{\sum_{i,j=1}^N w_{ij}^D} \quad (3)$$

where w_{ij}^D is the element in the i th row and j th column of the directed weight matrix, W .

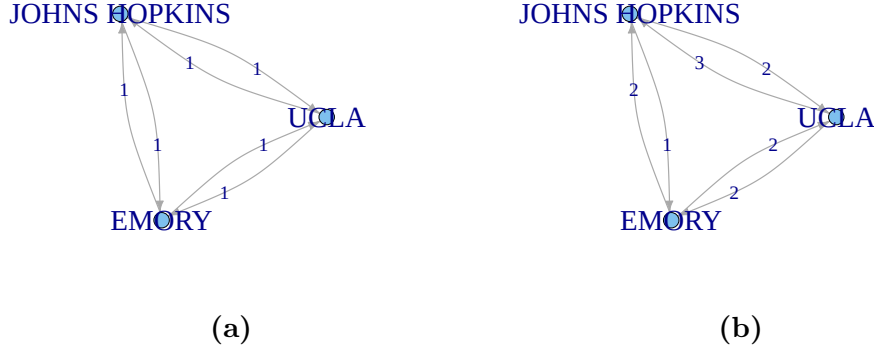


Figure 9: Visualization of a small subset of the NIH network used to assist in reciprocity illustration. The edge labels correspond to edge weights. (a) Unweighted network representation. (b) Weighted network representation.

The reason we take the minimum of w_{ij}^D and w_{ji}^D is to account for the fact that W is not symmetrical. The weighted reciprocity can also be thought of as project reciprocity. This will give us an idea of how many projects are being returned.

To further explain this notion of reciprocity, we consider Figure 9. When dealing with reciprocity in an unweighted network, like we see in Figure 9a, we can think of reciprocity in two different ways. The first is to think of the edges between organizations as being either bidirectional or not. In Figure 9a, we see the relationships between organizations are bidirectional, so the reciprocity is equal to one. The other way to think about it is in terms of the out-degrees of the organizations. Each organization has an out-degree of two and thus the total out-degree for this small network is six. Of those six ties, all six are being returned, so the reciprocity is equal to one. This second way of thinking about reciprocity in the unweighted network is the only way to correctly think about it in the weighted network.

Now, turning our attention to Figure 9b, we can see that the total out-degree is twelve. So, we can write $\sum_{i,j=1}^N w_{ij}^D = 12$. There are a total of three ties between Johns Hopkins and Emory: two going from Emory to Johns Hopkins and one going from Johns Hopkins to Emory. Of the ties from Johns Hopkins, one is being returned, so one is added to the numerator in the fraction in Equation (3). Of the edges from Emory, one is being returned,

so one more is added to the numerator. The denominator is going to gain three since two are extending from Johns Hopkins and one is extending from Emory. When applying this to all three nodes we see that the weighted reciprocity is equal to

$$rec_w = \frac{(1+1) + (2+2) + (2+2)}{(2+1) + (3+2) + (2+2)} = \frac{10}{12} = \frac{5}{6} = 0.833$$

The data show that, on an organizational level, the reciprocity is 100%, but on a project level, the reciprocity is only 83.3%.

5.8 Altruism

The next and final descriptive statistic is known as the level of altruism of a node. It is introduced by Wardil and Hauert [23]. It is calculated by taking the difference between the out-degree and in-degree and dividing the value by the sum of the out-degree and in-degree:

$$L_i = \frac{k_i^D - l_i^D}{k_i^D + l_i^D}$$

where k_i^D is the out-degree for node i and l_i^D is the in-degree for node i . An organization is called a pure egotist if its level of altruism, L_i , equals -1 and is called a pure altruist if $L_i = 1$. While these terms may be a bit too harsh or flattering, it is not difficult to image that being a secondary organization may be considered easier for the organization because it gets the benefits of funding and possibly publication without having to work on the project as long. That being said, this altruism level is a good measure of how an organization operates. If L_i is close to zero, the organization has an approximately equal number of instances in which it was primary and secondary.

To extend this to a directed and weighted network, replace out-degree and in-degree with

weighted out-degree and weighted in-degree.

$$L_i^W = \frac{s_i^D - l_i^{D,W}}{s_i^D + l_i^{D,W}}$$

where s_i^D is node strength for node i in a directed network and $l_i^{D,W}$ is the directed, weighted in-degree for node i .

6 NIH Network Data Analysis: Results

We will now present the results of all above descriptive statistics for all the network representations in question. Table 1 summarizes some results. We can see there is a large discrepancy between the out-degree for the undirected network representations versus the directed ones. This is partially due to the fact that we no longer count the in-degree. Only organizations that have been primary at some point will have out-degrees. Another reason that this difference is so large is due to the way the directed network representations were organized: with ties only going from primary to secondary organizations and no edges connecting secondary organizations that had worked on the same project. This table also reveals that the weighted network representations sheds light on new information. The weighted degrees (strengths) are higher than the unweighted degrees, which implies that many of the same organizations

Table 1: Descriptive Statistics for all four networks. UU is the undirected, unweighted network. UW is the undirected, weighted network. DU is the directed, unweighted network. DW is the directed, weighted network.

Network	UU	UW	DU	DW
Avg. out-degree (Strength)	13.52	18.49	5.00	6.00
Clustering Coefficient	0.480	0.491	0.187	0.215
Avg. Shortest Path	2.98	2.18	3.30	2.74
Avg. Closeness ($\times 10^{-3}$)	110.6	121.5	8.31	8.35
Avg. Betweenness ($\times 10^{-3}$)	4.25	5.81	2.67	3.08
Avg. Eigenvector Centrality	0.129	0.069	0.096	0.071

are working on more than just one grant together at some point.

Since we are using the inverse weights to find shortest paths for the weighted representations, the cost of traveling over an edge will be smaller if the node is more important. The fact that the average shortest paths for the weighted network representations is smaller than the average shortest paths for the unweighted implies that the shortest paths go through these well-connected organizations frequently. We also see quite a difference in the eigenvector centrality when comparing the unweighted network representations to their weighted counterparts. This suggests that there are many connections with smaller organizations that are not as well connected with the large ones.

We will now present the most important nodes in each network in terms of degree and betweenness. The degree, in Table 2, will tell how often an organization was involved in a collaborative project while the betweenness, in Table 3, will tell us how often it is used as a bridge between other organizations. One of the first things one might notice when observing these tables is the fact that different stories are told when we consider the different networks. Consider first Table 2. If one only considers how many different organizations an organization is connected to, the University of Pittsburgh would have the same ranking as the University of California San Diego (UCSD). However, if we also want to consider the number of times an

Table 2: Lists the five organizations in each network with the largest out-degree, in the case of unweighted networks, or largest weighted out-degree, in the case of the weighted networks.

Degree			
Undirected, Unweighted Network		Undirected, Weighted Network	
Organization	Degree	Organization	Weighted Degree
Johns Hopkins University	107	Johns Hopkins University	232
University of California San Diego	100	University of Pittsburgh	210
University of Pennsylvania	100	University of Pennsylvania	192
University of Pittsburgh	100	University of California Los Angeles	158
University of California Los Angeles	98	University of Alabama Birmingham	146
Directed, Unweighted Network		Directed, Weighted Network	
Organization	Degree	Organization	Weighted Degree
University of Pittsburgh	55	University of Pittsburgh	79
Johns Hopkins University	54	Johns Hopkins University	67
Washington University	51	Washington University	67
University of Texas Medical Center	47	Duke University	60
University of California San Diego	45	University of California San Diego	57

organization has been connected to others, then the University of Pittsburgh is clearly ranked higher than UCSD, which does not even appear in the top five organizations in this respect. Further, we can see that although Johns Hopkins has the most connections with different organizations, the University of Pittsburgh has, in fact, been the primary organization both with more individual schools and on more projects. It can also be seen that the top three schools do not change when we add weights to the directed network, which implies these organizations not only have been primary to the most other organizations, but also have been primary in the most individual projects.

Now turning our attention to Table 3, we see that the top five organizations for betweenness change for each network. For the undirected and unweighted network, Minnesota State Department of Health has a higher betweenness rating than any other organization. This is because, as seen in Figure 3, Minnesota State Department of Health links many other departments of health from other states across the country to different universities and medical organizations. However, since most of these state departments of health have small weights, Minnesota State Department of Health is not as important in the weighted networks and does not even appear in the top five for the directed networks despite the fact that 65% of its edges are reciprocated in the undirected, unweighted network.

Table 3: Lists the five organizations in each network with the largest betweenness centrality measure.

Betweenness			
Undirected, Unweighted Network		Undirected, Weighted Network	
Organization	Betweenness	Organization	Betweenness
Minnesota State Dept. of Health	0.080	University of Pittsburgh	0.307
University of Pittsburgh	0.079	Johns Hopkins University	0.254
University of Pennsylvania	0.075	University of Pennsylvania	0.131
University of California San Diego	0.068	Minnesota State Dept. of Health	0.105
Columbia University Health Sciences	0.062	Columbia University Health Sciences	0.076
Directed, Unweighted Network		Directed, Weighted Network	
Organization	Betweenness	Organization	Betweenness
University of Pittsburgh	0.054	University of Pittsburgh	0.109
Johns Hopkins University	0.046	Johns Hopkins University	0.088
Columbia University Health Sciences	0.043	Washington University	0.065
Washington University	0.043	Duke University	0.064
Duke University	0.042	Columbia University Health Sciences	0.056

Of course, where an organization wants to be in these tables differs. For instance, the University of Pennsylvania rubs shoulders with lots of other organizations but does not initiate as many relationships. In doing this, it takes part in many projects and gains information but does not impart information to others as often as organizations such as the University of Pittsburgh or Johns Hopkins University.

Minnesota State Department of Health, on the other hand, is not very important in terms of originating new ideas or new information, as evidenced by the fact that its out-degree is not very large. It is, however, very important in getting this information out to other organizations: other state departments of health. This is clear since its betweenness centrality is so high for the undirected networks.

While out-degree and betweenness provide insights into the fundamentals of the NIH network, they give much more preference to organizations that were involved with larger projects with many team members. It is also important to explore the number of different projects with which each organization was involved. Table 4 gives the list of the five organizations that were involved in the most individual projects. It is clear from this table that there are several organizations in the United States that have been much more active in terms of involvement in different projects. These organizations, namely the University of Pennsylvania, Johns Hopkins University and University of Pittsburgh, seem to be controlling the project direction for medical related research. Although they only comprise 0.66% of all the organizations in the network, these three organizations have been involved with around

Table 4: Lists the five organizations in the NIH network that were involved in the most projects.

Total Projects	
Organization	Projects
University of Pennsylvania	76
Johns Hopkins University	73
University of Pittsburgh	72
University of California Los Angeles	48
Washington University	48

16% of all the collaborative research projects in our database. It may be of great benefit for an organization to make collaborations at one, if not all three, of these organizations to contribute more to the cutting edge research projects that are being conducted.

Additionally, more than 39% of the organizations in this network were only involved with one project and around 54% were involved in two or less. The make-up of the NIH network is riddled with organizations that have only collaborated a handful of times. This could mean these organizations are not innovating as rapidly as the ones involved in more projects and are thus losing a competitive advantage when it comes to the amount of funding obtained from grants [24].

From the measures defined in the tables, it seems that the University of Pittsburgh is an important organization in the NIH Funding network. It is present in the top three organizations in nearly every category. The other seemingly important organizations include Johns Hopkins University, University of Pennsylvania, Washington University and Minnesota State Department of Health, where Minnesota State Department of Health is more important when it comes to betweenness and University of Pennsylvania is more involved with many different projects even though it is the primary organization in relatively few of them.

6.1 Reciprocity Results

Now we get into the reciprocity and altruism statistics. Both of these statistics only apply to the directed networks, so all network representations referenced here are directed. The reciprocity of the unweighted network representation, the organizational reciprocity, is equal to 0.4312. This means that about 43.12% of organizations that extended an edge to some other organization receive one back. Compare this with the weighted reciprocity, which is equal to 0.408. In other words, 40.8% of the number of projects are returned. So, on average, if organization i is the primary organization with organization j as the secondary organization on ten different grants, organization i can expect to be a secondary organization

4.1 times when organization j is primary, on average.

These reciprocity levels are considered to be quite high, which indicates that organizations tend to work with the same organizations quite a bit. It can be inferred that there are probably some strong ties between many organizations that collaborate with NIH funds. Of course, this may also mean that it could be difficult to join a project as a secondary organization because nearly half of the organizations will work with the same organization again. These results may be inflated a bit, however, since projects that have multiple primary organizations will have reciprocated edges. As mentioned earlier, projects with more than one primary organization account for 142 of the 1,230 projects in these networks.

6.2 Altruism Results

The altruism levels for these two directed networks will tell us which organizations are more likely to be primary organizations and which are more often secondary. The average level of altruism in the unweighted network is -0.221 and the average level of altruism in the weighted network is -0.218 . In both cases, the organizations take a little more than they give, on average. This makes intuitive sense since many projects have more secondary organizations than primary ones.

Looking at Figure 10, we can see the weighted out-degree of every organization in our weighted network representation plotted against their respective weighted altruism level. Notice that many of the organizations with high out-degrees tend to have an altruism level near zero. This informs us that these organizations are both passing on information and receiving information fairly evenly. The most interesting cases are shaded. Notice that Illinois State Department of Public Health has an altruism level of one, and yet, with an out-degree of twenty, it is in the top 8% in terms of out-degree. As is shown in Figure 6, Illinois State Department of Public Health is the primary organization in many projects that only contain other state departments of health and other public health organizations.

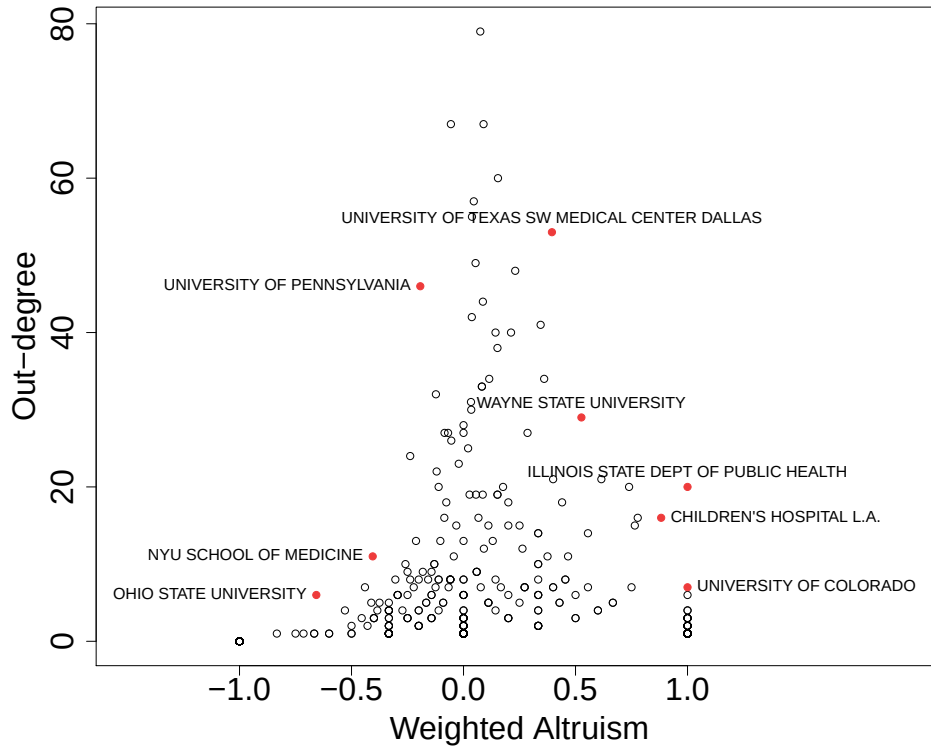


Figure 10: Weighted out-degree vs levels of altruism in the directed, weighted network. The shaded nodes are considered the most anomalous.

Most of these smaller health departments have never been a primary organization and so Illinois State Department of Public Health is acting as a vessel from which information flows, but this information only flows one way. There is a similar story with the University of Colorado at Boulder and the Children's Hospital of Los Angeles, which both have higher than average out-degrees yet have very high altruism levels as well. The opposite is true, although to a lesser extend, with Ohio State University and New York University School of Medicine. These schools both have out-degrees in the top 25% and top 15% respectively yet are secondary far more often than they are primary organizations.

The organizations with some of the highest out-degrees that are farther from zero than the other organizations with large out-degrees include the University of Pennsylvania and the University of Texas Southwestern Medical Center at Dallas. With an altruism level of -0.1930 and having been a secondary organization on twenty-two more projects than it has been a

primary organization, the University of Pennsylvania is certainly the most “egotistical” of the larger organizations. Again, being an egotist, is not necessarily a bad thing. This could mean that the primary organization was more likely to get funding or was able to obtain more funding if the University of Pennsylvania joined the team. The altruism level of the University of Texas Southwestern Medical Center at Dallas is 0.3947 and has been a primary organization on thirty more projects than for those on which it was a secondary organization.

Colorado State University’s weighted altruism level is 0.333, which means its out-degree is greater than its in-degree. However, CSU has worked on very few collaborative NIH grants so its total degree is only three, being the primary on two projects and being secondary once.

Figure 11 shows how the altruism levels change when going from unweighted to weighted. The six largest differences are shaded. Cleveland Clinic Foundation’s altruism level changes

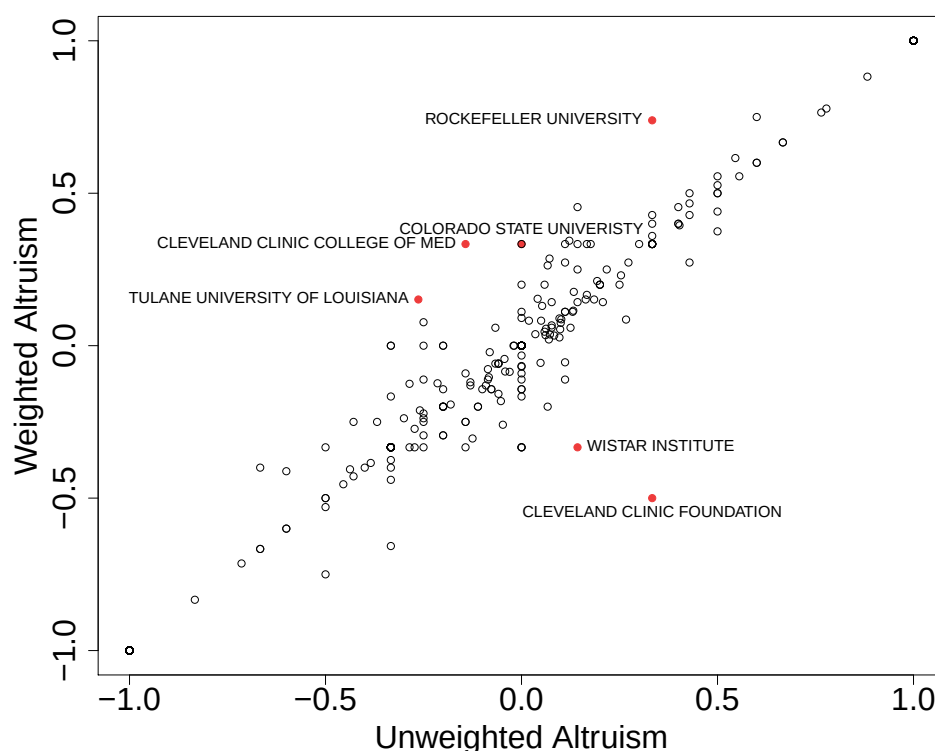


Figure 11: Levels of altruism in the directed, weighted network vs. the levels of altruism in the directed, unweighted network. The shaded nodes are those for which the two altruism levels differed the most.

the most of any organization in the network. If we look at just the unweighted, Cleveland Clinic Foundation’s altruism is 0.333. Then shifting over to the weighted altruism, this dropped considerably down to -0.5. So, it can be seen that organizations can be shed in a completely different light when only considering one out-degree per organization, rather than one out-degree per project. In this case, while Cleveland Clinic Foundation was the primary to two organizations and was the secondary to only one, it was secondary to that one on six different projects while it was the primary organization on only two distinct projects.

On the opposite side of the spectrum is Tulane University of Louisiana, whose unweighted altruism is negative but whose weighted altruism is positive. Colorado State University also has one of the largest changes in altruism. The unweighted altruism is zero since it was the primary organization to only one distinct organization and was a secondary organization once. However, when it was the primary organization, it had University of Nebraska Lincoln as its secondary on two different projects, so its weighted out-degree is two while its weighted in-degree is still only one. Therefore, its altruism rose from zero in the unweighted case to 0.333 in the weighted case.

7 Discussion

The analysis of social networks is definitely not trivial. What has been described in this paper is one way to do the analysis. We have seen that using weights equal to the number of different times organizations have worked together provided us with a different insight compared to just looking at the network with unweighted ties. Eventually, we will also want to weight the edges based on the amount of money that passed from the NIH to and between different organizations.

We also observed that the directed network told a different story from the undirected one. There are, however, different ways to define a primary organization, which will change the

structure of the network. Recall that an organization was considered primary first if it was involved with a certain project the most number of years and second if it was the starting organization. Another way to define a primary organization would be to first consider if it was the starting organization and then, if there were multiple starting organizations, consider the one that was involved the most number of years. Still further, one could define the primary organization as the one that received the most amount of money for that given project. All of these definitions would change the network structure, although we do not currently know how much.

A shortcoming of this study had to do with the NIH database itself. The fact that we found some project titles that were filed under slightly different names implies that there are more projects in the database that were not considered collaborative because of an entry mistake or difference in name when, in fact, they were. Something else to consider is the fact that the reciprocity and altruism level statistics could be slightly misleading since they do not distinguish between reciprocated edges being due to the fact that the two organizations both were primary on the same project or if they were both the sole primary on two different projects.

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